

Impact of urban environments on Vehicle-to-Vehicle (V2V) communication

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Abstract—Vehicle-to-Vehicle (V2V) communication is vital for autonomous safety, yet highly susceptible to urban environmental physics. This paper analyzes IEEE 802.11p DSRC signal degradation across diverse path-loss scenarios (Free Space, Two-Ray Ground, and Obstacle Shadowing) using the Veins simulation framework. Initial results showed idealized line-of-sight models achieving a 51.4% packet delivery ratio (PDR), while urban buildings plummeted the success rate to 20.2% by triggering over 116,000 PHY-layer signal drops. To isolate the root cause, a sensitivity analysis comparing a half-traffic urban grid against a straight-road highway was conducted. The data conclusively proves that severe signal attenuation from physical obstacle shadowing, rather than MAC-layer channel congestion, is the primary driver of network failure. These findings highlight the absolute necessity of employing realistic physical-layer analogue models when evaluating VANET protocols.

Index Terms—V2V, VANET, IEEE 802.11p, DSRC, OMNeT++, Veins, Obstacle Shadowing, Multipath Fading

I. INTRODUCTION

Cooperative Intelligent Transport Systems (C-ITS) rely heavily on the real-time exchange of safety beacons, such as Basic Safety Messages (BSMs), to prevent collisions and manage traffic flow. This communication is typically handled by the IEEE 802.11p standard operating in the 5.9 GHz frequency band.

Because 5.9 GHz radio waves require a near line-of-sight (LOS) to communicate effectively, the physical environment of a city becomes the greatest bottleneck to network reliability. Many early network simulations assumed vehicles communicate in a "vacuum" (Free Space), leading to overly optimistic results that fail in real-world deployment.

In this study, we transition from theoretical, idealized network models to highly realistic physical environments. We systematically isolate and measure the exact impact of asphalt reflections and concrete buildings on MAC-layer packet drop rates. By proving how the physical environment actively destroys data, we demonstrate the limits of standard V2V communication in dense urban grids.

II. RELATED WORK

The performance of Vehicular Ad-Hoc Networks (VANETs) has been the subject of extensive research, particularly regarding the limitations of the physical layer. The foundational guidelines for V2V communication are established by the IEEE 1609 family of standards and the 802.11p amendment

[1], which dictate the Medium Access Control (MAC) and Physical (PHY) layer behaviors for Dedicated Short-Range Communications (DSRC).

Building upon this, further research by Sommer et al. [2] introduced the computationally efficient "Obstacle Shadowing Model" for OMNeT++, which we directly utilize in this study. This model departs from overly optimistic free-space assumptions and accurately calculates radio wave attenuation based on the specific intersections of signals with simulated concrete building polygons. By utilizing these established frameworks, our study directly builds upon prior efforts to bring realistic multipath fading and physical signal absorption into VANET evaluation.

III. PROBLEM DEFINITION

To accurately model the physical network, we utilized a bi-directionally coupled simulator: SUMO (Simulation of Urban MObility) to handle realistic vehicle traffic physics, and OMNeT++ to handle the radio network stack. We assume that all vehicles are broadcasting standard safety beacons at regular intervals.

To isolate the environmental impact, we tested three specific analogue models at the Physical Layer (PHY):

- 1) **Free Space:** Assumes an empty void where signal strength decays purely over distance ($1/d^2$).
- 2) **Two-Ray Ground:** Introduces the asphalt street. The receiving antenna calculates the direct radio wave *plus* the radio wave bouncing off the ground, causing violent signal fluctuations (multipath fading).
- 3) **Obstacle Shadowing:** Introduces physical polygons (buildings) into the simulation. If a line-of-sight intersects a building wall, the signal is aggressively attenuated (absorbed).

A. Baseline Simulation Parameters

The exact configuration used for the baseline OMNeT++ simulations is detailed below:

- **Spawning Time:** 1 car per second
- **Simulation Time:** 100 seconds
- **Playground Size:** 600m x 600m
- **MAC Protocol:** IEEE 1609.4 / 802.11p
- **Transmitter Power:** 20 mW
- **Bitrate:** 6 Mbps
- **Frequency:** 5.89 GHz (DSRC band)
- **Mobility Model:** Veins (TraCI coupled with SUMO)

A more comprehensive visualization of the baseline simulation's playground is shown in Figure 1 below:

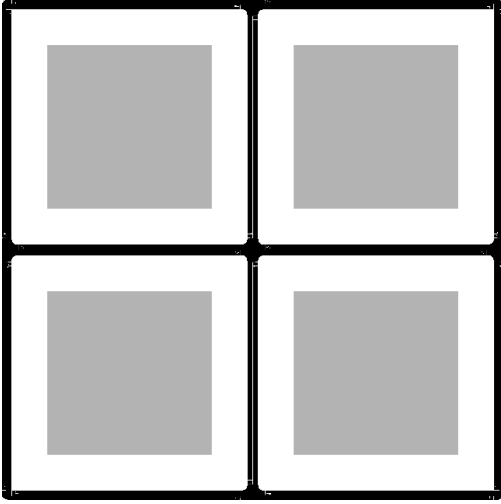


Fig. 1. Simulation's playground: Urban grid with vehicle nodes and building polygons.

B. Sensitivity Control Scenarios

To rigorously isolate whether network degradation is caused by MAC-layer interference or PHY-layer attenuation, two specialized control scenarios were introduced, modifying the baseline parameters:

- **Half Traffic Scenario:** The urban grid environment (600m x 600m) is maintained with building obstacles, but the vehicle spawn rate is reduced from 1 vehicle/s to 1 vehicle every 2 seconds. This halves the total vehicle count to 50, explicitly reducing channel congestion.
- **Straight Road Scenario:** Obstacle attenuation is completely removed by placing vehicles on a single 1-kilometer highway without building polygons. To accommodate the extended road, the OMNeT++ playground is expanded to 1100m x 200m. The original spawn rate (1 vehicle/s) is maintained to preserve high traffic density in a purely line-of-sight environment.

A more comprehensive visualization of the *Straight Road* simulation's playground is shown in Figure 2 below:

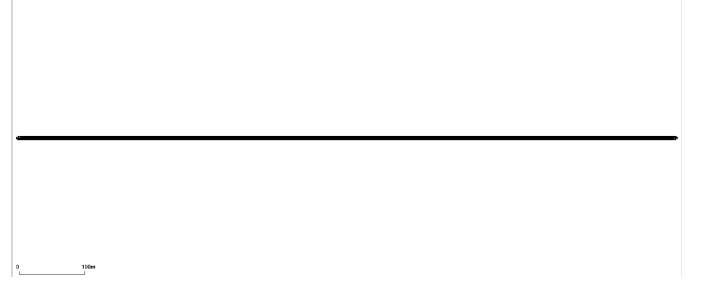


Fig. 2. Straight Road playground: 1-kilometer highway without buildings

IV. RESULTS

To ensure the statistical validity of the results and avoid anomalies caused by specific pseudorandom number generator (RNG) seeds, each of the three environmental scenarios was simulated using **10 independent replications**. The OMNeT++ engine was configured to use a different RNG seed for mobility and network traffic generation in each run.

The data extracted from these 30 total simulations was aggregated to calculate the statistical mean for each metric. Furthermore, **95% Confidence Intervals (CI)** were computed using the Student's t-distribution to provide strict error margins, ensuring a scientifically fair comparison between the Free Space, Two-Ray Ground, and Urban Shadowing models.

A. Traffic Density Validation

Before analyzing the radio network performance, it is crucial to validate the physical traffic density generated by SUMO. Because vehicles dynamically enter and exit the simulation map upon reaching their individual destinations, the total number of *concurrent* transmitting vehicles at any given second differs from the absolute spawn count.

To accurately correlate channel congestion with physical density, the mean maximum number of concurrent vehicles was extracted across all 10 replications for each scenario.

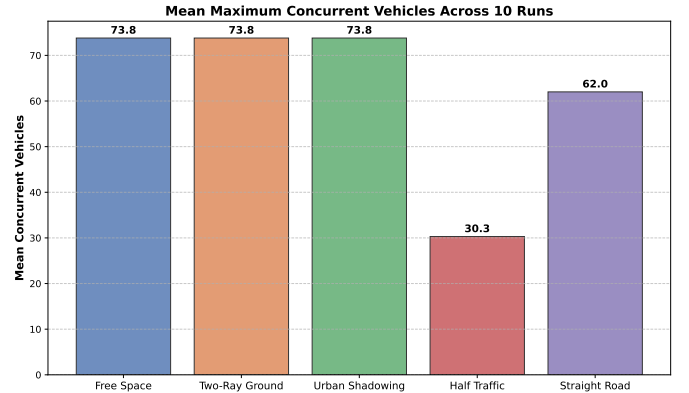


Fig. 3. Mean maximum concurrent vehicles active on the map across 10 independent replications.

As illustrated in Figure 3, the baseline urban scenarios (Free Space, Two-Ray Ground, and Urban Shadowing) reached a mean maximum of approximately **73.8** concurrent vehicles. By reducing the spawn rate in the Half Traffic scenario, the concurrent vehicle peak was successfully halved to **30.3**, validating our control setup for reducing MAC-layer congestion. Finally, the extended Straight Road scenario maintained a peak of **62.0** concurrent vehicles. This confirms that the mobility models executed exactly as intended, providing a stable and proven physical foundation for the subsequent radio network analysis.

B. Packet Delivery Ratio (PDR)

The Packet Delivery Ratio is the primary metric for evaluating the reliability of a VANET. Because IEEE 802.11p safety messages (WSMs and BSMs) are transmitted as unacknowledged broadcasts, the theoretical expected receptions were calculated as the number of broadcasts multiplied by the number of potential receivers in the network.

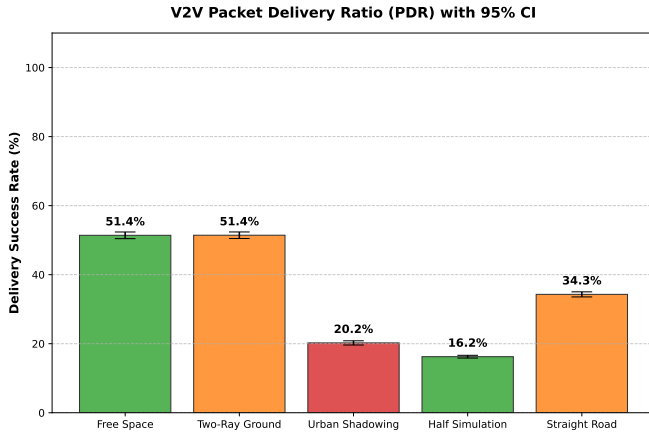


Fig. 4. Packet Delivery Ratio across environments with 95% Confidence Intervals.

As shown in Figure 4, the Free Space and Two-Ray Ground models exhibit similar PDRs (51.4%). However, the introduction of physical buildings in the Urban Shadowing scenario causes a catastrophic collapse in network reliability, plummeting the PDR to just 20.2%. This proves that idealized line-of-sight path-loss formulas are dangerously inadequate for modeling real-world city deployments.

C. Channel Congestion and PHY-Layer Interference

A common misconception in VANET simulations is analyzing standard MAC-layer packet drops to measure congestion. However, because IEEE 802.11p broadcasts do not utilize Acknowledgements (ACKs) or MAC-layer retries, in-air collisions are actually handled by the Physical (PHY) layer. When two overlapping transmissions arrive at a receiver, the PHY layer perceives the overlap as destructive noise, heavily degrading the Signal-to-Noise plus Interference Ratio (SNIR) and causing the packet to be dropped before it ever reaches the MAC layer.

Therefore, to accurately measure MAC-layer congestion, we must analyze how often the CSMA/CA protocol forces a vehicle to delay transmission (Backoff). To measure actual signal destruction, we analyze the PHY-layer SNIR drops.

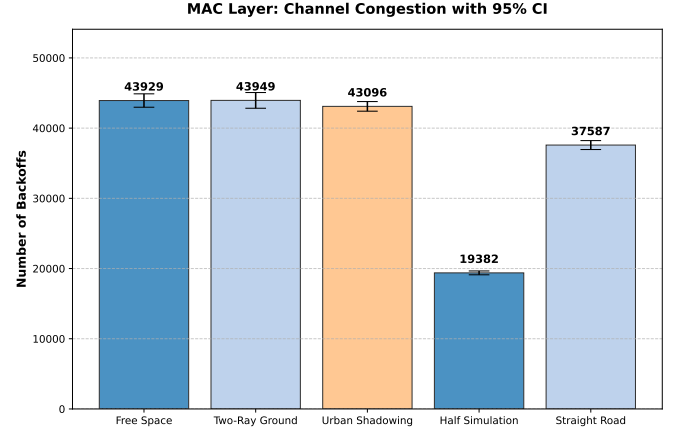


Fig. 5. MAC Layer Channel Congestion: Times forced into backoff.

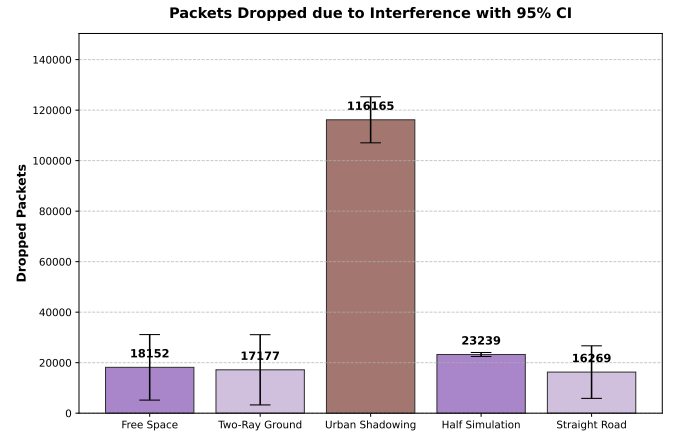


Fig. 6. Packets destroyed at the PHY layer due to poor SNIR and interference.

As illustrated in Figure 5, channel congestion remains relatively stable across environments, as vehicles correctly utilize CSMA/CA to back off when the channel is busy. However, Figure 6 reveals the true cause of the Urban Shadowing PDR collapse. While the Free Space environment suffers only ~18,000 drops due to standard distance attenuation, the Urban Shadowing model triggers a massive spike of over 116,000 SNIR drops. The building polygons cause severe multipath fading, signal absorption, and non-line-of-sight (NLOS) dead zones, physically destroying the radio waves in the air.

D. Sensitivity Analysis: Congestion vs. Attenuation

Following the initial findings, a sensitivity analysis was conducted using the two control scenarios defined in Section III-B. Specifically, we sought to verify whether the performance collapse was strictly due to signal attenuation (buildings physically blocking radio waves) or artificially inflated channel

interference (high vehicle density causing in-air collisions). Both control scenarios were simulated over 10 independent replications.

The data definitively proves that signal attenuation, not channel congestion, is the primary cause of network failure in urban environments.

In the **Half Traffic** scenario, MAC-layer channel congestion dropped massively (from 43,096 backoffs down to just 19,382, as seen in Figure 5). Despite this drastically cleaner channel, the overall PDR remained critically low at 16.2% (Figure 4). This confirms that reducing interference does not salvage the network; the buildings physically absorb the signals before they can reach receiving antennas around the corner.

Conversely, in the **Straight Road** scenario, removing the buildings caused the SNIR drops to plummet back to baseline levels (16,269 drops, mirroring Free Space). Because the line-of-sight was restored, the PDR immediately recovered to 34.3%. The delivery rate naturally plateaus before reaching 100% due to the 1-kilometer highway exceeding the maximum physical transmission range (approx. 300 meters).

Ultimately, this confirms that in dense urban grids, the concrete buildings act as physical isolators. While they inadvertently reduce widespread channel interference by blocking overlapping signals, they actively destroy the vital line-of-sight connections required for 5.9 GHz V2V communication.

V. CONCLUSION

This study evaluated the impact of environmental physics on IEEE 802.11p V2V networks through bidirectionally coupled simulations using SUMO and OMNeT++. Initial baseline tests proved that network reliability is intrinsically tied to physical surroundings: idealized line-of-sight models successfully delivered 51.4% of packets, whereas the addition of urban buildings caused a network collapse, plummeting the delivery success rate to just 20.2%.

To directly address whether this failure was caused by high-density channel interference or physical signal absorption, a sensitivity analysis was conducted. Halving the vehicle traffic successfully reduced MAC-layer congestion by over 50%, yet the PDR remained critically low at 16.2%. Conversely, placing the vehicles on a straight highway completely removed obstacle shadowing, causing SNIR drops to plummet and restoring the PDR to 34.3%.

Ultimately, this data confirms that concrete buildings act as physical isolators. While they inadvertently reduce widespread channel interference by blocking overlapping signals, they actively destroy the vital line-of-sight connections required for 5.9 GHz V2V communication. Future urban VANET designs must account for severe obstacle shadowing, potentially by relying on edge infrastructure such as Roadside Units (RSUs) placed at intersections, to relay messages around blind corners.

REFERENCES

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- [2] C. Sommer, D. Eckhoff, R. German, and F. Dressler, “A computationally inexpensive empirical model of IEEE 802.11p radio shadowing in urban environments,” in 2011 Eighth International Conference on Wireless On-Demand Network Systems and Services, Jan. 2011, pp. 84-90.